DMM: Disparity-Guided Multispectral Mamba for Oriented Object Detection in Remote Sensing

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***Abstract*— Multispectral-oriented object detection faces challenges due to both intermodal and intramodal discrepancies. Recent studies often rely on transformer-based models to address these issues and achieve cross-modal fusion detection. However, the quadratic computational complexity of transformers limits their performance in remote sensing imagery. Inspired by the efficiency and lower complexity of Mamba in long-sequence tasks, we propose disparity-guided multispectral Mamba (DMM), a multispectral-oriented object detection framework comprised of a disparity-guided cross-modal fusion Mamba (DCFM) module, a multiscale target-aware attention (MTA) module, and a target-prior aware (TPA) auxiliary task. The DCFM module leverages disparity information between modalities to adaptively merge features from RGB and infrared (IR) images, mitigating intermodal conflicts. The MTA module aims to enhance feature representation by focusing on relevant target regions within the RGB modality, addressing intramodal variations. The TPA auxiliary task utilizes single-modal labels to guide the optimization of the MTA module, ensuring it focuses on targets and their local context. Extensive experiments on the DroneVehicle and VEDAI datasets demonstrate the effectiveness of our method, which outperforms state-of-the-art methods while maintaining computational efficiency. Code will be available at https://github.com/Another-0/DMM**

***Index Terms*— Multispectral object detection, oriented object detection, remote sensing, state-space model (SSM).**

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I. Introduction

OBJECT detection is a fundamental task in the field of remote sensing, with extensive applications in civilian and military areas such as urban planning, traffic surveillance, disaster relief, and military reconnaissance [1], [2], [3], [4]. However, traditional object detection methods often struggle with the unique challenges posed by remote sensing images [6], [7], such as high variability in scale, orientation, and density of objects. To address these challenges, oriented object detection has been developed, which includes the detection of object angles. This approach significantly enhances the accuracy and precision of detecting rotated, irregular, and densely packed objects in complex remote sensing environments [8]. Consequently, oriented object detection has garnered considerable research attention and application [9], [10], [11], [12], [13]. However, these studies are predominantly designed for normal visible (RGB) images, which often suffer from sparse information and significant noise interference due to low illumination and complex weather conditions.

With advancements in optical sensor technology, infrared (IR) images have been widely adopted as an additional modality to address these challenges, as they can stably reflect the thermal information and contours of objects regardless of illumination and weather [14], [15]. Although IR images are less impacted by variations in illumination and weather conditions, they offer fewer details about objects, such as color and texture. Therefore, it is intuitive to combine RGB and IR images for object detection, a technique known as multispectral object detection [16], [17].

Based on previous studies [5], [14], [18], [19], [20] and our observations, the challenges for multispectral object detection in remote sensing images can be categorized into two types. One challenge is the intermodal differences caused by modal characteristics, shooting angles, calibration errors, and postprocessing. Fig. 1(a) provides some examples of these intermodal disparities. Another challenge is the substantial intramodal differences within RGB images, where the image quality of targets can vary dramatically under different lighting

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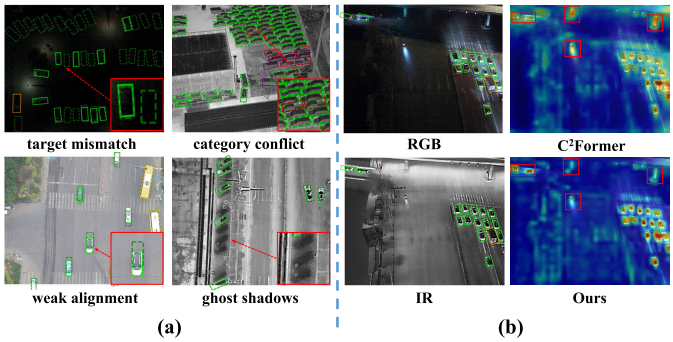


Fig. 1. Modality disparities between RGB and IR images. Different color bounding boxes represent different categories. Bounding boxes from the RGB modality are represented with solid lines, while dashed lines are from IR modality. (a) Examples of intermodal disparities. Target mismatch arises from the varying visibility of targets across different modalities. Category conflict indicates that the differences cause confusion in manual annotation. Due to calibration errors of the capturing devices, paired images are not perfectly aligned. The characteristic of IR thermal crossover imaging results in ghost shadows in the IR images. (b) Intramodal disparities in RGB images. Uneven illumination can generate a lot of misleading information. C^2Former [5] incorrectly focuses on these anomalous regions, whereas our method suppresses background noise and pays more attention to the regions of interest with targets.

conditions, as shown in Fig. 1(b). Specifically, artificial light sources at night, such as streetlights, car headlights, and neon signs, can generate misleading information. Additionally, severe noise and shadows caused by extremely low illumination and underexposure interfere with feature extraction, leading to the inclusion of a large amount of irrelevant information in the RGB modality, which hinders optimal fusion in subsequent steps.

Recent studies have applied convolutional neural networks (CNNs) and Transformers to address the fusion challenges. TSFADet [19] designed a CNN-based alignment framework to solve the weak alignment problem in paired image modalities. However, due to the fixed receptive field size, CNN-based methods struggle to learn global contextual information, which is crucial for detecting numerous small targets in remote sensing images. While Transformer [21] has shown effectiveness in capturing long-range dependencies and global information, their expensive computational burden limits its application to high-resolution remote sensing images [22]. Consequently, hybrid methods combining CNNs and Transformers have been proposed. For instance, C^2Former [5] uses CNNs to extract image features and then employs an intermodality cross-attention module based on the Transformer architecture to obtain aligned and complementary features, addressing the issue of inaccurate cross-modal fusion. Despite the encouraging results of these methods, they still cannot avoid the quadratic computational complexity of Transformers. Additionally, to reduce computational load, these methods often project features to lower dimensions when using Transformers to compute global attention, which inevitably compromises fusion efficiency. Moreover, existing methods have not adequately explored the differences within the RGB modality and coupled these with intermodal discrepancies, which are crucial for effective multispectral object detection, particularly under varying illumination conditions.

To address the aforementioned problems, we propose a disparity-guided multispectral Mamba (DMM) oriented object detection framework called DMM. It includes a disparity-guided cross-modal fusion Mamba (DCFM) module to address intermodal differences and a multiscale target-aware attention (MTA) module to handle intramodal differences within the RGB modality. Benefiting from the selective scanning mechanism and hardware-friendly efficient computation strategies of Mamba [23], [24], DCFM models the global interaction awareness of both single-modality information and modality difference information without sacrificing computational efficiency. MTA employs multiscale convolutional windows to extract both target and local background information from RGB features, aiming to suppress invalid interference within the modality. Additionally, to guide MTA learn the effective information, a target-prior aware (TPA) auxiliary task is designed, which introduces additional supervisory information through pseudo-labels or manual annotations, using a pretrained auxiliary detection head to constrain MTA's optimization process, thus obtaining more beneficial RGB features for subsequent fusion.

Our contributions can be summarized as follows.

1) We propose a novel multispectral-oriented object detection framework, DMM, based on the Mamba architecture. Leveraging Mamba's ability to efficiently capture long-range dependencies, the DCFM module of DMM fuses cross-modal features guided by modality disparity information effectively. To the best of authors' knowledge, DMM represents the first successful application of Mamba for multispectral-oriented object detection.
2) We develop the MTA module along with an auxiliary task TPA to bridge the feature gap within the RGB modality. TPA introduces additional supervisory information, enabling MTA to focus more on regions where targets are present.
3) We validate the effectiveness of our proposed method through extensive experiments conducted on two widely used remote sensing datasets. Our method improves mAP@0.5 by 5.2% on the DroneVehicle dataset and by 5.8% on the VEDAI dataset, surpassing existing state-of-the-art methods and setting a new benchmark for multispectral oriented object detection.

The rest of this article is structured as follows. Section II provides a brief overview of related works. Section III introduces the proposed method, and we conduct extensive experiments to validate its effectiveness in Section IV. Finally, we conclude this work in Section V.

II. Related Work

A. Oriented Object Detection

To extend general object detection methods to rotational scenarios, Yang and Yan [10] and Yang et al. [25] modeled angle prediction as a classification problem, whereas the majority of research focuses on regression-based methods. Rotated RPN [26], RoI Transformer [27], and Oriented R-CNN [28] propose various strategies to improve the quality of anchor generation for classical anchor-based object detectors. Inspired

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by ATSS [29], methods such as DAL [30], SASM [31], GGHL [32], and Oriented Reppoints [33] explore the impact of label assignment strategies on oriented object detection performance from different perspectives. KFIoU [34] proposes an effective approximation of SkewIoU based on Gaussian modeling, and PKINet [35] addresses the challenges of large object scale variations and complex backgrounds in remote sensing scenes by improving the feature extraction network. To balance accuracy and speed, single-stage-based refined detectors, such as R^3Det [11] and S^2A-Net [36], have been proposed. Although these methods have achieved encouraging results in general daytime scenarios, their performance is limited in low-light conditions due to reliance solely on the RGB modality.

B. Multispectral Object Detection

Introducing the IR modality in oriented object detection provides robustness against illumination variations [37], [38], [39]. Numerous studies focus on effectively fusing data from both the RGB and IR modalities [40], [41]. Halfway Fusion [42] pioneeringly demonstrated that feature-level fusion yields better results compared to other fusion levels. AR-CNN [18] addressed weak alignment discrepancies in multimodal fusion by designing a region feature alignment (RFA) module to capture positional shifts and adaptively align regional features of both modalities. SuperYOLO [43] fused RGB and IR images at the input level, leveraging a super-resolution branch to enhance feature representation and LF-MDet [44] improved the efficiency of multimodal object detection through frequency filtering and low-rank matrix decomposition, but they failed to account for the influence of object orientation. TSFADet [19] proposed an alignment module to predict the deviation between the two modalities and calibrate the feature maps. The latest method C^2Former [5] designed an intermodal cross-attention (ICA) module to address both modality calibration and fusion inaccuracies and an adaptive feature sampling (AFS) module to balance the computational cost of global attention. They implemented their modules based on transformer and incorporate numerous strategies to reduce computational load at the expense of fusion accuracy. This prompts us to consider whether it is possible to achieve superior fusion detection results without compromising computational efficiency.

C. Mamba

Mamba is an enhancement of state-space models (SSMs) [46], [47], [48], [49] that introduces a selection mechanism and a hardware-aware algorithm, enabling parameterization of SSMs based on the input sequence for efficient processing of long sequences in discrete modalities [23], [24]. Recent advances in Mamba have shown significant potential in efficiently handling long-sequence modeling [50], [51], [52], [53]. Mamba offers an alternative to the attention-based models like Transformers by scaling linearly with sequence length and effectively modeling long-range dependencies [51], [53]. Efforts like Vision Mamba [54] and VMamba [45] extend Mamba's capability to process visual data through bidirectional and multidirectional scanning approaches. Notably, Mamba has also been widely applied in remote sensing tasks [55], [56], [57]. RSMamba [56] incorporates a dynamic multipath activation mechanism to enhance the modeling of noncausal data and demonstrates superior performance in remote sensing image classification. Pan-Mamba [58] leverages the Mamba model's efficiency in global information modeling for pan-sharpening, incorporating channel swapping and cross-modal Mamba components to achieve superior fusion results. Samba [59] employs a unique encoder architecture, based on the Mamba design, to effectively extract semantic information from high-resolution remotely sensed images. Despite the existing research on applying Mamba to general multimodal tasks, its application in multispectral oriented object detection remains unexplored.

III. Method

This section introduces the proposed disparity-guided multispectral Mamba detection framework, as shown in Fig. 2, which consists of three main components: a dual-stream feature extraction backbone network based on a 2-D selective SSM, a MTA module guided by TPA auxiliary task, and a cross-modal feature fusion module guided by modal disparity (DCFM). We will first introduce the fundamental concepts of the visual selective SSM in Section III-A. Then, in Section III-B, we will describe the proposed DCFM module. Section III-C will cover the proposed MTA and TPA modules and their joint optimization process. Finally, in Section III-D, we will present the entire oriented object detection process with the integrated detection head and discuss our loss function.

A. Preliminaries

1) State-Space Models: SSMs are a class of models that represent the internal state of a system as a set of linear equations. They are particularly useful for modeling dynamic systems and can be used to capture temporal dependencies in data. Specifically, SSMs can be formulated as linear time-invariant (LTI) systems, which process a 1-D input sequence $x_t \in \mathbb{R}$ by passing it through intermediate implicit states $h_t \in \mathbb{R}^N$ to produce an output $y_t \in \mathbb{R}$, where N is the dimension of the hidden layer. These LTI systems are governed by the following equations and characterized by a set of parameters: the state transition matrix $\mathbf{A} \in \mathbb{R}^{N\times N}$, the projection parameters $\mathbf{B} \in \mathbb{R}^{N\times 1}$, $\mathbf{C} \in \mathbb{R}^{1\times N}$, and the skip connection $D \in \mathbb{C}^1$:

$$\begin{aligned} h_t' &= \mathbf{A}h_t + \mathbf{B}x_t \\ y_t &= \mathbf{C}h_t + Dx_t. \end{aligned} \tag{1}$$

Due to the continuous-time nature of SSMs, directly applying them in the field of deep learning poses significant challenges, as discretization is required. In the context of computer vision, this means that the continuous SSM must be converted into a discrete form to be compatible with neural network architectures typically used for tasks such as

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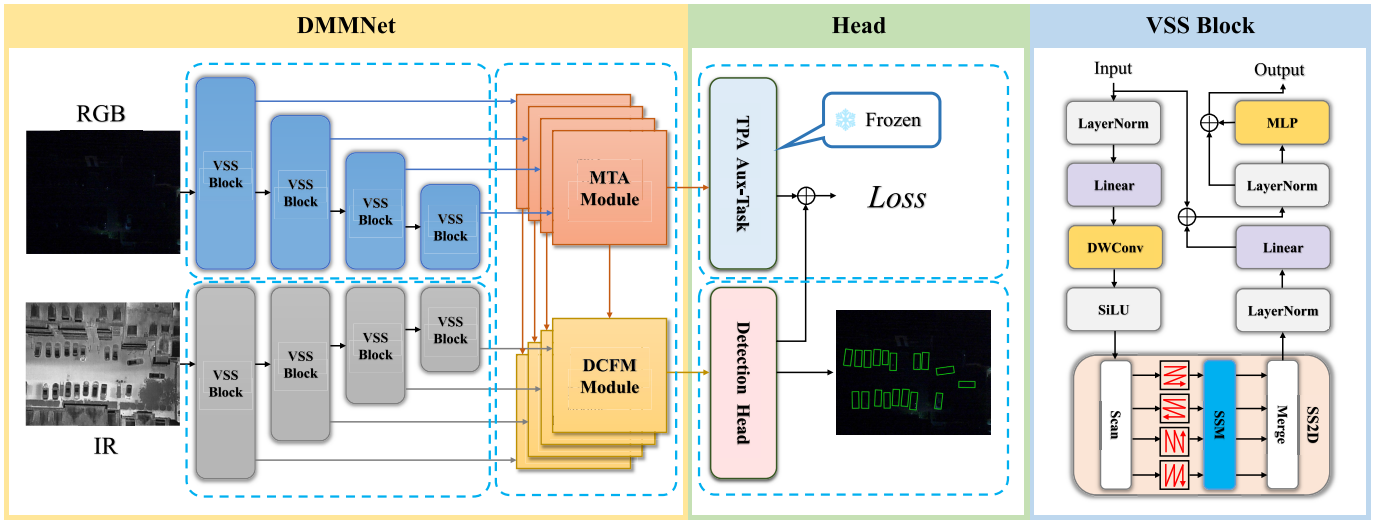


Fig. 2. Overview of our proposed DMM method. The input dual-modal features are first projected into a high-dimensional space through convolution, followed by feature extraction using VSS blocks. Each VSS block is cascaded with a downsampling layer to reduce the feature map size. The features of different sizes generated by the VSS blocks in the upper stream and lower stream are fed into the MTA module and the DCFM module, respectively. The output of the MTA is fed into the TPA head to assess the quality of the MTA-enhanced features. The output of the DCFM is directed to the subsequent detection head. Given the critical role of the FPN structure in both one-stage and two-stage algorithms, it is included as part of the detection head architecture. On the far right, we present the structure of the VSS block within the backbone and the SS2D mechanism at the lowest module level, derived from the V9 architecture of VMamba [45].

image recognition, object detection, and segmentation. The discretization process involves sampling the continuous-time model at specific intervals, leading to a discrete-time approximation. This transformation is essential for integrating SSMs with standard deep learning frameworks, which operate on discrete data points. Specifically, the discretization of SSMs can be achieved by solving the ODE in (1), using methods similar to the zero-order hold principle [23]. For a timescale parameter Δ, the formulas for converting the continuous parameters $\mathbf{A}$ and $\mathbf{B}$ into their discrete counterparts $\bar{\mathbf{A}}$ and $\bar{\mathbf{B}}$ are as follows:

$$\begin{aligned}\bar{\mathbf{A}} &= \exp(\Delta\mathbf{A}) \\ \bar{\mathbf{B}} &= (\Delta\mathbf{A})^{-1}(\exp(\Delta\mathbf{A}) - \mathbf{I}) \cdot \Delta\mathbf{B}.\end{aligned} \tag{2}$$

Therefore, (1) can be rewritten as

$$\begin{aligned}h_t &= \bar{\mathbf{A}}h_{t-1} + \bar{\mathbf{B}}x_t \\ y_t &= \bar{\mathbf{C}}h_t + Dx_t \\ \bar{\mathbf{C}} &= \mathbf{C}\end{aligned} \tag{3}$$

where D is regarded as a residual connection and is typically omitted in the equation.

2) Selective-Scan Mechanism: Traditional SSMs, while effective for discrete sequences, face challenges due to their invariant parameters. The selective SSM (S6), also known as Mamba, overcomes this by deriving the matrices $\mathbf{B}$, $\mathbf{C}$, and Δ from the input data x_t, thus making the model contextually aware [23]. The selective scan mechanism represents a significant advancement in the modeling of complex sequences by introducing input-dependent parameters, addressing the limitations of traditional LTI systems. In computer vision, the direct application of Mamba is challenged by the inherent differences between 2-D visual data and 1-D sequences. Vision tasks require capturing spatial information, which is not the primary focus in 1-D sequence modeling. To bridge this gap, a 2-D selective scan (SS2D) mechanism is introduced in VMamba [45]. SS2D arranges image patches in four directions, as shown in Fig. 2, where the directions represent the manner in which the 2-D feature map is flattened into a 1-D sequence, generating separate sequences that integrate information from all directions, thereby establishing a global receptive field. This method ensures comprehensive feature integration without increasing computational complexity. The SS2D mechanism processes each directionally scanned sequence with the S6 model, combining the resultant features to reconstruct the 2-D feature map. This approach allows each element in the 1-D array to interact with previously scanned samples through a condensed hidden state, effectively capturing contextual information across the entire image.

B. Disparity-Guided Cross-Modal Fusion Mamba

The feature fusion module is the core of multimodal models. Existing effective methods primarily apply the dynamic modeling capabilities of transformers to address intermodal differences. However, their expensive computational cost needs careful consideration. Unlike previous approaches, our DCFM module is based on Mamba, achieving global attention while avoiding quadratic computational overhead. The structure of our proposed DCFM module is shown in Fig. 3. Considering a set of inputs Feat_{rgb}, $\text{Feat}_{\text{ir}} \in \mathbb{R}^{N\times C\times H\times W}$ representing RGB and IR features, respectively, we first normalize the features using a LayerNorm block to accelerate model convergence. Then, we calculate the intermodal disparities and project all of them to a unified hidden space

$$\begin{aligned}F_{\text{rgb}} &= \text{LN}(\text{Feat}_{\text{rgb}}) \\ F_{\text{ir}} &= \text{LN}(\text{Feat}_{\text{ir}}) \\ F_d &= F_{\text{rgb}} - F_{\text{ir}} \\ F_i' &= \text{Project}(F_i), \quad F_i = F_{\text{rgb}}, F_{\text{ir}}, F_d\end{aligned} \tag{4}$$

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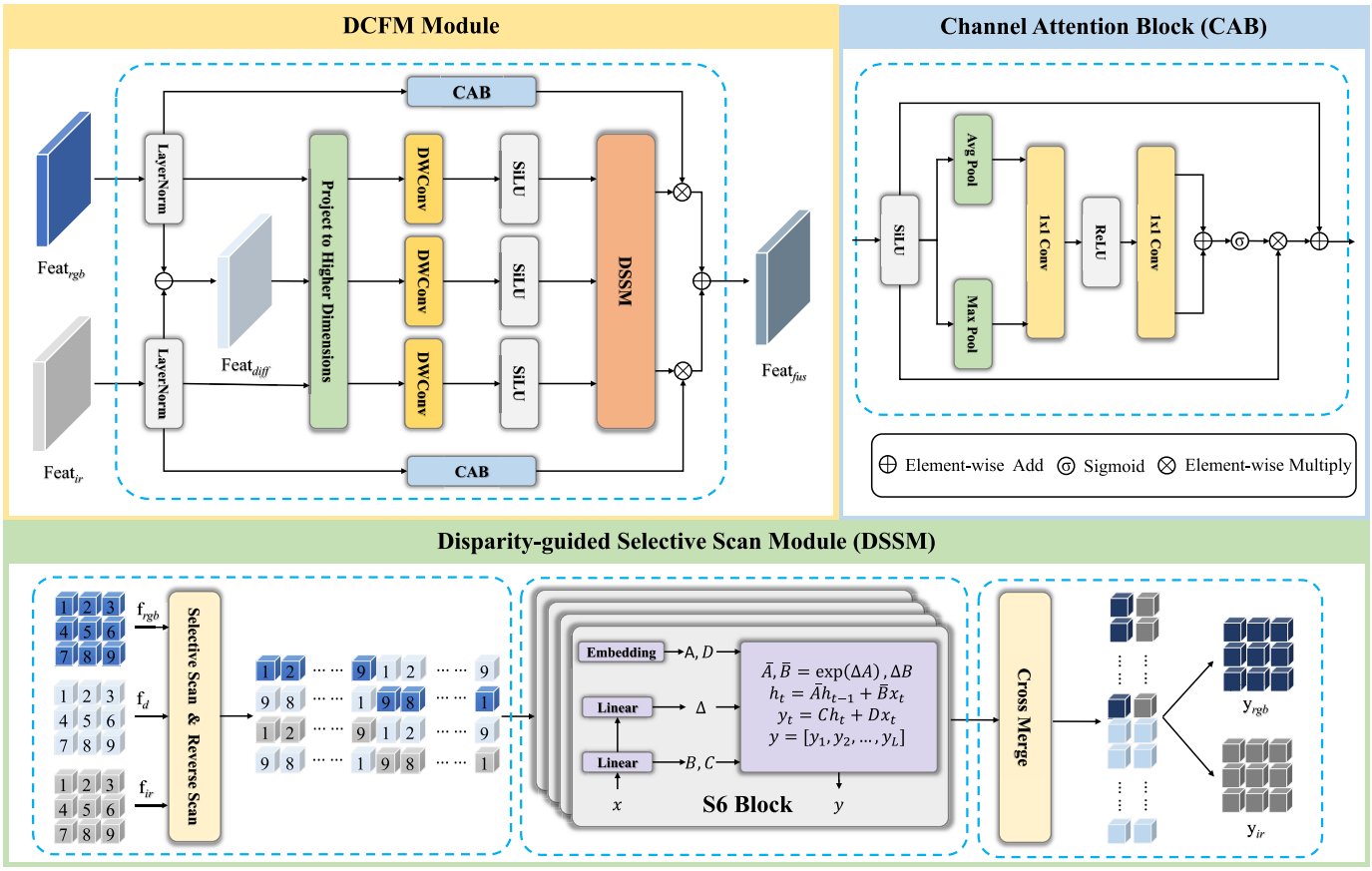


Fig. 3. Structures of our proposed DCFM module. The DCFM module projects RGB and IR features to higher dimensions and combines them using DSSM. The CAB enhances feature representation and the DSSM refines and merges features.

where LN(·) represents the LayerNorm operation and Project(·) represents the projection operation using a linear transformation. Unlike the common dimensionality reduction operations in transformers, the channel dimension of $X' \in \mathbb{R}^{N\times C'\times H\times W}$ is typically twice that of the initial dimension ($C' = 2C$), providing better modality information representation. Subsequently, depthwise separable convolutions DWConv(·) are applied to the three features to facilitate interchannel communication

$$f_i = \mathrm{SiLU}\big(\mathrm{DWConv}\big(F_i'\big)\big), \quad F_i' = F_{\mathrm{rgb}}', F_{\mathrm{ir}}', F_d' \tag{5}$$

where SiLU(·) represents the SiLU activation function. These features, fully integrated with information from all channels, undergo further processing through the disparity-guided selective scan module (DSSM) module. This module extracts complementary information from another modality while suppressing redundant data. The out features are then elementwise multiplied and summed with the original features processed by the channel attention block (CAB), ultimately yielding high-quality fused features.

1) Channel Attention Block: To explicitly model the interchannel relationships, we propose CAB, which computes channel attention on the normalized original features, adaptively learning intramodal channel interactions to enhance the feature representation of a single modality. Specifically, for a given input feature F_i, we compute the channel attention as follows:

$$\begin{aligned} F_{\mathrm{avg}} &= \mathrm{AvgPool}(\mathrm{SiLU}(F_i)) \\ F_{\mathrm{max}} &= \mathrm{MaxPool}(\mathrm{SiLU}(F_i)) \\ w &= \mathcal{S}\big(\mathrm{CR}(F_{\mathrm{avg}}) + \mathrm{CR}(F_{\mathrm{max}})\big) \\ F_{\mathrm{out}} &= w \cdot F_i + F_i \end{aligned} \tag{6}$$

where CR(·) denotes the operation of two 1×1 convolutions with an ReLU activation function applied between them, and $\mathcal{S}(\cdot)$ represents the Sigmoid function. Both AvgPool(·) and MaxPool(·) are global operations, producing F_{avg} and F_{max} in $\mathbb{R}^{N\times C\times 1\times 1}$.

2) Disparity-Guided Selective Scan Module: The unique selective scan mechanism in the Mamba architecture allows it to adjust parameters selectively based on the input data. Leveraging this context-aware capability, we designed the DSSM module. Specifically, the DSSM processes three inputs, f_{rgb}, f_{ir}, and $f_d \in \mathbb{R}^{N\times D\times H\times W}$. First, we flatten the features and then concatenate f_d with f_{rgb} and f_{ir} to obtain two input sequences $f_{\text{rgb-}d}$ and $f_{\text{ir-}d} \in \mathbb{R}^{N\times 2HW\times D}$. Considering the 2-D nature of image features, we reverse scan the sequences to obtain two new input sequences, $\overline{f}_{\text{rgb-}d}$ and $\overline{f}_{\text{ir-}d}$. These four sequences are then processed using (2) and (3) to produce output sequences $f'_{\text{rgb-}d}$, $f'_{\text{ir-}d}$, $\overline{f}'_{\text{rgb-}d}$, and $\overline{f}'_{\text{ir-}d} \in \mathbb{R}^{N\times 2HW\times D}$. Finally, the outputs are fed into the cross-merge module, where the two inverted sequences are reversed and added back to their corresponding sequences. The computation is performed

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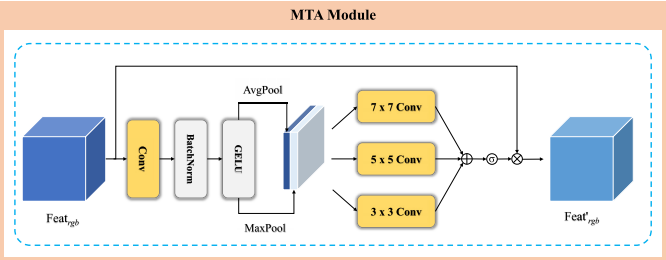


Fig. 4. Structure of our proposed MTA module. The outputs are fed to both the DCFM module for high-precision fusion and the TPA module for loss calculation.

according to the following formula:

$$y_{\text{rgb-}d} = f'_{\text{rgb-}d} + \text{Reverse}\left(\overline{f}'_{\text{rgb-}d}\right)$$
$$y_{\text{ir-}d} = f'_{\text{ir-}d} + \text{Reverse}\left(\overline{f}'_{\text{ir-}d}\right) \tag{7}$$

where Reverse$(\cdot)$ function reverses the input along the second dimension. We, then, perform the inverse operation of concatenation by splitting $y_{\text{rgb-}d}$ and $y_{\text{ir-}d}$ along the same dimension. Retaining the modality-specific features from the first half and discarding the differential features from the second half, we reshape them into the final output features $y_{\text{rgb-}d}$ and $y_{\text{ir-}d} \in \mathbb{R}^{N\times D\times H\times W}$.

C. Multiscale Target-Aware Attention

Although the DCFM module effectively integrates multimodal information, enhancing the representation and understanding of complex visual data, intramodal variations such as lighting can introduce interference and redundancy in RGB features, as shown in Fig. 1(b). This interference hinders the fusion module's ability to effectively distinguish between relevant targets and background noise, thus compromising fusion performance. To address this issue, we propose the MTA module, as depicted in Fig. 4. The MTA module computes multiscale spatial self-attention to adaptively focus on target regions within the RGB modality, thereby providing higher quality RGB features for subsequent fusion modules.

Specifically, for a single-layer feature $x_{\text{rgb}} \in \mathbb{R}^{N\times C\times H\times W}$ output by the VSS block, we first apply a convolution layer for initial processing. We then use channelwise AvgPool$(\cdot)$ and MaxPool$(\cdot)$ operations to aggregate channel information. Finally, we compute global spatial attention weights using convolution operations with different kernel sizes. These weights are used to reweight the original input features, which are then combined with a residual connection to produce the enhanced features.

TPA Auxiliary Task: From our observation, the MTA module alone is insufficient to ensure that the reweighted features focus more on the targets and suppress background noise. To address this, we introduce the TPA auxiliary task by adding a TPA head after the MTA module. The structure of TPA is similar to the FPN and RPN layers in traditional single-stream object detectors, but with the parameters frozen during training. We first obtain the TPA weights by pretraining an RGB single-modal two-stage object detector, using either pseudo-labels generated by other SOTA single-modal detectors or manually annotated labels. Then, we pretrain a detector that includes only the DCFM module to enable the model to learn robust representations from dual-modal images. Finally, we incorporate the MTA and TPA modules into the model for joint training. During the training of our model, the feature outputs by the MTA module are simultaneously fed to the TPA detection head, as shown in Fig. 2. The RPN layer's role in extracting regions of interest aligns with our optimization goal, so we use the RPN loss as the optimization target for the auxiliary task. With the TPA auxiliary task constraint, the MTA module is optimized to focus on target regions identified by the RPN, resulting in higher quality RGB features for subsequent fusion modules. The effectiveness of this strategy is clearly illustrated in Section IV-C through ablation experiments.

D. Dual-Stream Detector

As described earlier, our DMM is a dual-stream feature extraction and fusion network that operates independently of the subsequent detection head. This means that our model can be integrated into any existing SOTA object detection methods simply by replacing the feature extraction architecture. In this article, we adopt the one-stage method S^2A-Net [36] as our basic detector, adhering to the same configurations as previous SOTA models. To facilitate experimental comparisons and verify generalization, we also integrate our method into the classic two-stage methods Faster R-CNN [60] and Oriented R-CNN [28] by replacing the feature extraction module. All the detectors use oriented bounding box (OBB) detection heads. We retain their unique detection head designs without any modifications to ensure fair comparisons with other methods.

Loss Function: We adhere to the design of other detection networks, retaining their inherent loss functions. The distinction lies in augmenting these with the loss functions from our TPA auxiliary task. Therefore, the final loss function is defined as follows:

$$\text{Loss}_{\text{All}} = \text{Loss}_{\text{det_cls}} + \text{Loss}_{\text{det_reg}} + \text{Loss}_{\text{aux_cls}} + \text{Loss}_{\text{aux_reg}} \tag{8}$$

where $\text{Loss}_{\text{det_cls}}$ and $\text{Loss}_{\text{det_reg}}$ represent the classification and regression losses of the detection head, respectively, while $\text{Loss}_{\text{aux_cls}}$ and $\text{Loss}_{\text{aux_reg}}$ denote the classification and regression losses of the TPA auxiliary task. All classification loss functions use cross-entropy loss [61], with binary classification for the TPA auxiliary task; all regression losses use smooth L_1 loss [62] function. For two-stage detection algorithms, the first two terms include both the RPN network losses and the ROI network losses.

IV. Experiments

A. Implementation Details

We train and validate our model on the following two datasets.

1) *DroneVehicle Dataset:* The DroneVehicle dataset [14] is a large-scale remote sensing dataset collected from drones, comprising 28 439 pairs of RGB and IR images.

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Each image is annotated with OBBs for five categories (car, bus, truck, van, and freight car), totaling 953 087 instances. The scenes range from daytime to nighttime, including roads, urban areas, and parking lots. The dataset is officially divided into training, validation, and test sets with 17 990, 1469, and 8980 image pairs, respectively. During training, we remove the 100-pixel-wide white borders around all images and conduct comparative experiments on the test sets.

2) *VEDAI Dataset:* The VEDAI dataset [63] is designed for vehicle detection in high-resolution aerial images. It includes diverse urban and rural scenes with oriented annotations for various vehicle types such as cars, trucks, and vans, along with a few other objects like airplanes and ships, totaling nine categories. The dataset consists of 1246 pairs of RGB and IR images with resolutions of 1024×1024 and 512×512 pixels. In our experiments, we use the 512×512 version and follow the tenfold cross-validation protocol as described in [63] to train and test our model.

3) *Experimental Settings:* Our experiments were conducted on an NVIDIA 4090 GPU with 24 GB of memory. The model's code environment is based on CUDA 11.6 and PyTorch 1.13.1, built upon the MMdetection and MMrotate frameworks. We employed pretrained VMamba as our backbone network, with the input image resolution set to the dataset's native resolution, consistent with the compared methods, and a batch size of 2. The optimizer used is AdamW, with an initial learning rate of 0.0001 and a weight decay of 0.05. Data augmentation is limited to random flipping with a probability of 0.5. To enhance model stability, we removed all zero-area bounding boxes from the ground truths during the training step. The model was trained for 12 epochs to ensure a fair comparison with previous methods. For the DroneVehicle dataset, we followed the settings from previous studies [5], [14], [19], using the IR modality labels as ground truth. For the VEDAI dataset, we trained the model on all nine categories, including those with fewer than 50 instances.

4) *Metric:* We adopt the mean average precision (mAP) as the evaluation criterion. mAP is computed as the arithmetic mean of the AP values across all categories. The AP value is obtained for each category separately by computing the average of interpolated precision $P(R)$ at recall levels $R \in \{0.0, 0.1, 0.2, \ldots, 1.0\}$. For each recall level R, the interpolated precision is defined as the maximum precision observed for recall values greater than or equal to R, ensuring that fluctuations in the precision–recall curve do not negatively impact the final evaluation. The AP is then computed across all 11 recall levels as follows:

$$\mathrm{AP} = \frac{1}{11} \sum_{n=0}^{10} \max_{R \geq \frac{n}{10}} P(R) \tag{9}$$

where $P(R)$ represents the precision as a function of recall.

We adopt both mAP@0.5 and mAP@0.5:0.95 for evaluation. The mAP@0.5 corresponds to the PASCAL VOC criterion where a predicted bounding box is considered correct if the intersection over union (IoU) with the ground-truth box is at least 0.5. The mAP@0.5:0.95 follows the COCO-style metric, which averages AP values over ten IoU thresholds ranging from 0.5 to 0.95 with a step size of 0.05, providing a more comprehensive assessment of detection accuracy under stricter localization constraints.

B. Comparisons With State-of-the-Art Methods

1) Comparisons on DroneVehicle Dataset: Table I presents the comparative results of our method against competing approaches. The compared methods include classic object detection approaches based on single-modal RGB and IR data: RetinaNet, R3Net, S^2ANet, and KFIoU for one-stage methods; and Faster R-CNN, RoITransformer, Oriented R-CNN, and PKINet-S for two-stage methods. Additionally, we also compare with multimodal fusion detection methods such as Halfway Fusion, AR-CNN, CIAN, MBNet, TSFADet, and C^2Former. It is evident that most of multimodal fusion detection methods significantly outperform the single-modal detectors, demonstrating the substantial advantages of multimodal data over single-modal data.

In the previous multimodal detectors, C^2Former achieves the highest detection accuracy with 74.2% mAP@0.5, followed by TSFADet with 73.9% mAP@0.5. By contrast, our method achieves the highest mAP@0.5 of 79.4%, surpassing the SOTA by 5.2 points. Under the more rigorous evaluation metric of mAP@0.5:0.95, our method also achieves the highest score, further validating its superiority over existing approaches. Furthermore, when integrating our proposed modules into other classic detectors like Faster R-CNN, we also observed superior performance exceeding the current SOTA. Additionally, Table I reveals that the performance improvement of our method is primarily due to enhanced detection capabilities for categories with fewer instances, such as Truck (8657), Freight-Car (5064), and Van (4282). We provide visual comparisons of detection results on DroneVehicle dataset in Fig. 5. As illustrated by the blue dashed circles in the figure, our method identifies more instances with greater accuracy in challenging scenarios such as foggy weather and nighttime compared to SOTA methods, which aligns with our experimental observations. These comprehensive comparative experiments demonstrate the effectiveness and generalization capability of our method.

2) Comparisons on VEDAI Dataset: We integrated our DMM into S^2A-Net and Oriented R-CNN, Table II presents our experimental results on the VEDAI dataset, and Fig. 6 provides visual comparisons of detection results. Currently, mainstream multimodal fusion detection research on the VEDAI dataset is primarily based on horizontal bounding box (HBB), with limited exploration of OBB. Therefore, we mainly compare our approach with single-modal detectors on the VEDAI dataset, such as the one-stage methods RetinaNet and S^2A-Net, and the two-stage methods Faster R-CNN, RoITransfomer, and Oriented R-CNN. For multimodal methods, we compared with the convolutional attention-based

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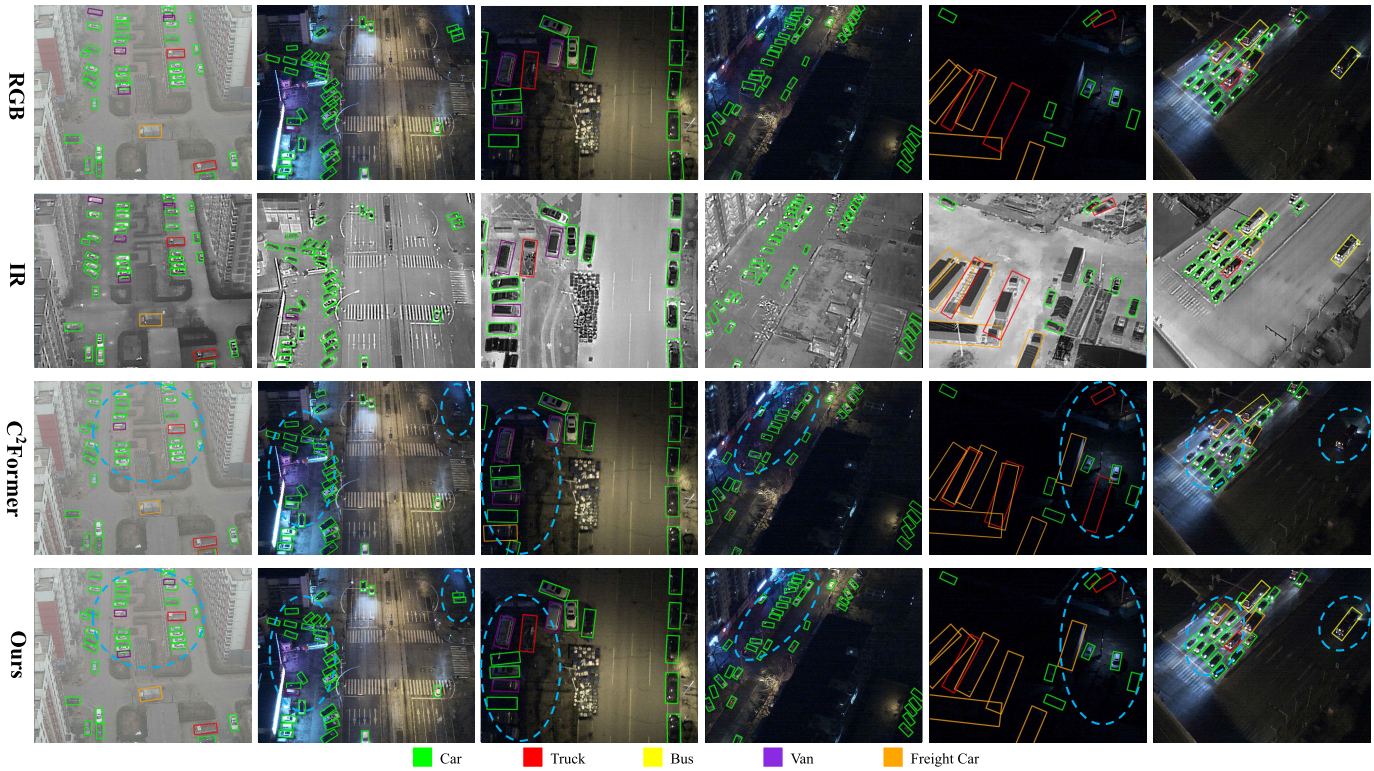


Fig. 5. Visualization of prediction results on the DroneVehicle dataset, with a confidence threshold set to 0.5. The detection boxes in RGB and IR represent ground truths. The comparison of detection results within the blue dashed circles indicates that our method demonstrates superior visual performance for each category.

TABLE I

COMPREHENSIVE COMPARATIVE EXPERIMENTS ON THE DRONEVEHICLE DATASET. WE COMPARED THE DMM METHOD WITH BOTH SINGLE-MODAL AND MULTISPECTRAL OBJECT DETECTORS, ALL EMPLOYING OBB DETECTION HEADS. THE BEST RESULTS ARE HIGHLIGHTED IN BOLD, WHILE THE SECOND-BEST RESULTS ARE UNDERLINED

Method	Basic Detector	Type	Modality	Car	Truck	Freight Car	Bus	Van	mAP@0.5:0.95	mAP@0.5
RetinaNet [64]	-	One Stage	RGB	78.5	34.4	24.1	69.8	28.8	25.0	47.1
R^3Det [11]	-	One Stage		80.3	56.1	42.7	80.2	44.4	27.4	60.8
S^2ANet [36]	-	One Stage		80.0	54.2	42.2	84.9	43.8	31.4	61.0
KFIoU [34]	-	One Stage		76.3	50.0	33.2	78.2	35.7	28.0	54.7
Faster R-CNN [60]	-	Two Stage		79.0	49.0	37.2	77.0	37.0	28.5	55.9
RoITransformer [9]	-	Two Stage		61.6	55.1	42.3	85.5	44.8	32.9	61.6
Oriented R-CNN [28]	-	Two Stage		80.1	53.8	41.6	85.4	43.3	32.7	60.8
PKINet-S [35]	-	Two Stage		76.4	53.0	40.8	79.1	45.7	31.2	59.0
RetinaNet [64]	-	One Stage	IR	88.8	35.4	39.5	76.5	32.1	30.4	54.5
R^3Det [11]	-	One Stage		89.5	48.3	16.6	87.1	39.9	36.7	62.3
S^2ANet [36]	-	One Stage		89.9	54.5	55.8	88.9	48.4	40.4	67.5
KFIoU [34]	-	One Stage		90.2	65.0	48.5	89.0	43.0	41.2	67.1
Faster R-CNN [60]	-	Two Stage		89.4	53.5	48.3	87.0	42.6	40.1	64.2
RoITransformer [9]	-	Two Stage		89.6	51.0	53.4	88.9	44.5	41.2	65.5
Oriented R-CNN [28]	-	Two Stage		89.8	57.4	53.1	89.3	45.4	41.5	67.0
PKINet-S [35]	-	Two Stage		90.2	67.3	56.3	88.8	51.2	43.8	70.8
UA-CMDet [14]	RoITransformer	Two Stage	RGB+IR	87.5	60.7	46.8	87.1	38.0	-	64.0
Halfway Fusion [42]	Faster R-CNN	Two Stage		90.1	62.3	58.5	89.1	49.8	42.9	70.0
CIAN [37]	-	One Stage		90.1	63.8	60.7	89.1	50.3	-	70.8
AR-CNN [18]	Faster R-CNN	Two Stage		90.1	64.8	62.1	89.4	51.5	-	71.6
MBNet [65]	-	One Stage		90.1	64.4	62.4	88.8	53.6	-	71.9
TSFADet [19]	Oriented R-CNN	Two Stage		89.9	67.9	63.7	89.8	54.0	-	73.1
TSFADet [19]	Cacade R-CNN	Two Stage		90.0	69.2	65.5	89.7	55.2	-	73.9
C^2Former [5]	S^2ANet	One Stage		90.2	68.3	64.4	89.8	58.5	47.5	74.2
DMM(Ours)	Faster R-CNN	Two Stage	RGB+IR	**90.4**	77.8	63.0	88.7	66.0	48.1	77.2
DMM(Ours)	S^2ANet	One Stage		**90.4**	**79.8**	**68.2**	**89.9**	**68.6**	**52.1**	**79.4**

method CMAFF and the SOTA method C^2Former on the DroneVehicle dataset.

From the experimental results, it can be observed that two-stage methods significantly outperform one-stage meth-

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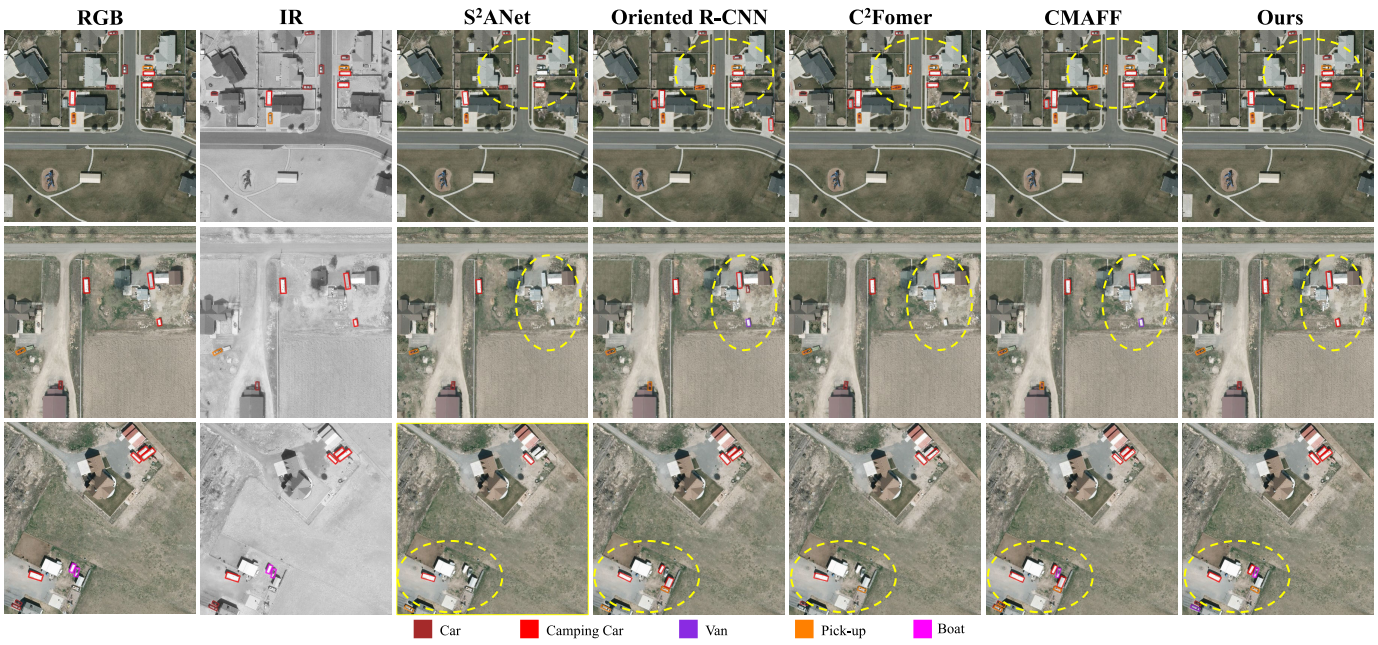


Fig. 6. Visualization of prediction results on the VEDAI dataset, with a confidence threshold set to 0.5. The detection boxes in RGB and IR represent ground truths. The comparison of detection results within the yellow dashed circles demonstrates that our method has a distinct advantage in identifying small aerial targets.

TABLE II

COMPARATIVE EXPERIMENTS ON THE VEDAI DATASET. ALL THE METHODS UTILIZE OBB DETECTION HEADS AND THE METRICS ARE THE AVERAGES FROM TENFOLD CROSS-VALIDATION. THE BEST RESULTS ARE HIGHLIGHTED IN BOLD, WHILE THE SECOND-BEST RESULTS ARE UNDERLINED

Method	Type	Modality	Car	Truck	Tractor	Camping Car	Van	Pick-up	Boat	Plane	Others	mAP@0.5:0.95	mAP@0.5
RetinaNet [64]	One Stage	RGB	48.9	16.8	15.9	21.4	5.9	37.5	4.4	21.2	14.1	8.5	20.7
S^2ANet [36]	One Stage		74.5	47.3	55.6	61.7	32.5	65.1	16.7	7.1	39.8	18.9	44.5
Faster R-CNN [60]	Two Stage		71.4	54.2	61.0	70.5	59.5	67.6	52.3	77.1	40.1	24.3	61.5
RoITransformer [9]	Two Stage		77.3	56.1	64.7	73.6	60.2	71.5	56.7	85.7	42.8	27.6	65.4
Oriented R-CNN [28]	Two Stage		77.6	59.7	62.8	76.7	60.9	72.3	60.1	84.0	43.6	29.4	66.4
RetinaNet [64]	One Stage	IR	44.2	15.3	9.4	17.1	7.2	32.1	4.0	33.4	5.7	8.0	18.7
S^2ANet [36]	One Stage		73.0	39.2	41.9	59.2	32.3	65.6	13.9	12.0	23.1	17.2	40.0
Faster R-CNN [60]	Two Stage		71.6	49.1	49.2	68.1	57.0	66.5	35.6	71.6	29.5	21.6	55.4
RoITransformer [9]	Two Stage		76.1	51.7	51.9	71.2	64.3	70.7	46.9	83.3	28.3	24.6	60.5
Oriented R-CNN [28]	Two Stage		77.0	55.0	47.5	73.6	63.2	72.2	49.4	79.6	30.5	27.1	60.9
C^2Former+S^2ANet [5]	One Stage	RGB+IR	76.7	52.0	59.8	63.2	48.0	68.7	43.3	47.0	41.9	22.9	55.6
DMM+S^2ANet(Ours)	One Stage		77.9	59.3	68.1	70.8	57.4	75.8	61.2	77.5	43.5	28.1	65.7
CMAFF+Oriented R-CNN [20]	Two Stage		81.7	58.8	68.7	78.4	68.5	76.3	66.0	72.7	51.5	30.5	69.2
DMM+Oriented R-CNN(Ours)	Two Stage		**84.2**	**65.7**	**72.3**	**79.0**	**72.5**	**78.8**	**72.3**	**93.6**	**56.2**	**39.7**	**75.0**

ods. This is primarily because the instance sizes in the VEDAI dataset are very small, and except for several vehicle categories, other categories are inconsistently distributed across different fold splits, making it challenging for one-stage methods to directly regress to the precise location of the targets. Consequently, C^2Former achieved only 55.6% mAP@0.5 and 22.9% mAP@0.5:0.95 on the VEDAI dataset, whereas our DMM + S^2A-Net achieved 65.7% mAP@0.5 and 28.1% mAP@0.5:0.95, even surpassing most two-stage single-modal methods. Additionally, the CMAFF method based on Oriented R-CNN achieved 69.2% mAP@0.5 and 30.5% mAP@0.5:0.95, higher than all single-modal methods, while our DMM + Oriented R-CNN exceeded it by 5.8% points in mAP@0.5 and 9.2% points in mAP@0.5:0.95. The experimental results fully demonstrate the generality of our method for oriented object detection tasks in remote sensing.

C. Ablation Study

We conducted our ablation experiments on the DroneVehicle dataset. To eliminate the influence of different detectors, we adopted the classic Faster R-CNN as the baseline and modified it into a dual-stream-oriented detector. As shown in Table III, incorporating the Mamba model resulted in a 5.3% increase in mAP@0.5. Adding the DCFM module further increased the mAP@0.5 by 0.5%, demonstrating the effectiveness of the Mamba model in processing remote sensing images. However, directly using the MTA module led to a 0.2% decrease in mAP@0.5. In contrast, introducing the TPA auxiliary task raised the mAP@0.5 to 76.1%. Integrating all the modules, the model achieved its best performance with an mAP@0.5 of 77.2%. Additionally, the design of the detection head also influences model performance. Combining DMM with the high-performance S^2A-Net further improved

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TABLE III

ABLATION EXPERIMENTS ON THE DRONEVEHICLE DATASET. F REPRESENTS THE FASTER R-CNN AND S REPRESENTS THE S^{2}ANET

Method	Detector		Module				mAP@0.5
	F	S	Mamba	DCFM	MTA	TPA	
Baseline(B)	✓	-	-	-	-	-	70.5
B+Mamba	✓	-	✓	-	-	-	75.8
B+Mamba+DCFM	✓	-	✓	✓	-	-	76.3
B+Mamba+MTA	✓	-	✓	-	✓	-	75.6
B+Mamba+MTA+TPA	✓	-	✓	-	✓	✓	76.1
DMM+Faster R-CNN	✓	-	✓	✓	✓	✓	77.2
DMM+S^{2}ANet	-	✓	✓	✓	✓	✓	**79.4**

TABLE IV

COMPARISON OF OUR DMM METHOD WITH THE LATEST SOTA METHOD C^{2}FORMER IN TERMS OF THE NUMBER OF MODEL PARAMETERS, THE MAXIMUM IMAGE SIZE THAT CAN BE PROCESSED DURING INFERENCE ON A 24-GB NVIDIA 4090 GPU, AND DETECTION PERFORMANCE

Method	PARAMS	MAXIMUM INPUT SIZE	mAP@0.5
C^{2}Former	118.47M	1092x1092	74.2
DMM	87.97M(30.5M↓)	3016x3016(762.8%↑)	79.4(5.2↑)

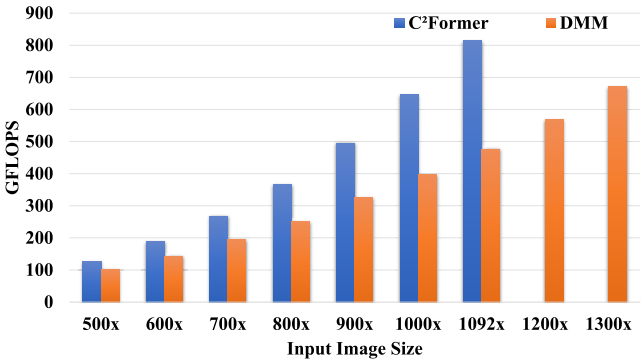


Fig. 7. Comparison of GFLOPs for the model with different input image sizes. The input images have the same width and height, so only one dimension is marked on the horizontal axis. Due to the quadratic complexity of C^{2}Former, its memory usage exceeds physical limits for input sizes larger than 1100 × 1100; hence, they are not shown.

the mAP@0.5 to 79.4%. These results indicate that each component contributes to the overall improvement in mAP@0.5.

D. Comparisons of Computational Costs

Table IV presents a comparison between our method and the previous SOTA method C^{2}Former. Both approaches use S^{2}ANet as the base detector and adhere to the same configuration. However, our model not only maintains a lower parameter count, with just 87.97M parameters, reducing approximately 25%, but also achieves the highest detection performance. Moreover, as shown in Fig. 7, with increasing input image sizes, the giga floating-point operations per second (GFLOPs) of our model increase almost linearly. In contrast, C^{2}Former's GFLOPs, though initially close to our model for smaller image sizes, significantly surpasses ours for larger images due to the quadratic complexity of the Transformer. Furthermore, with larger input image sizes, the memory usage of C^{2}Former increases significantly. On a 24-GB NVIDIA 4090 GPU, C^{2}Former supports a maximum inference size of only 1092 × 1092, whereas our model can handle inputs up to 3016 × 3016, which is approximately 7.6 times larger. These experimental results demonstrate the superior capability of our model in handling high-resolution images in the remote sensing domain.

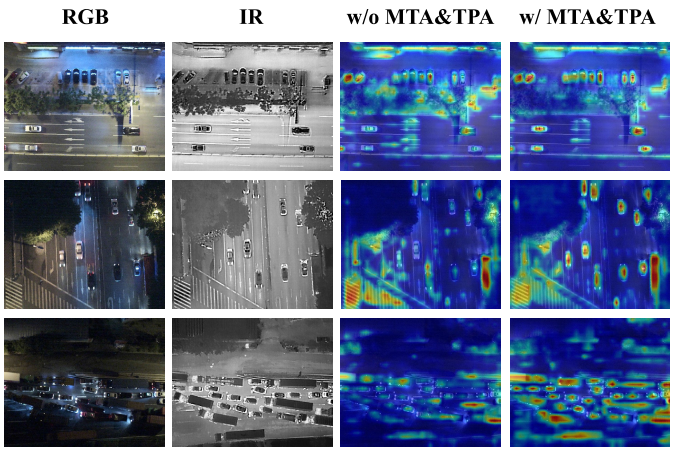


Fig. 8. Visualization of intermediate feature maps. Comparison of the fused features with and without the use of the MTA and TPA modules.

E. Feature Analysis

To validate the effectiveness of our method in enhancing detection performance, we present a comparative visualization of intermediate feature maps in Fig. 8. The comparisons show feature maps with and without these modules. It is evident that with the incorporation of the MTA and TPA modules, the model exhibits a higher degree of focus on the target objects while effectively suppressing background noise. To quantitatively demonstrate the effectiveness, we propose a metric called spatial feature attention contrast (SFAC). SFAC is defined as the ratio of the mean feature intensity within the ground-truth bounding boxes to that of the background, where feature intensity is derived from the attention maps. Specifically, given an attention map, we first upsample it to match the original image size and normalize the values to the range [0, 255]. Let $\mathbf{I}_k(i, j)$ denote the intensity of pixel (i, j) in the kth image. The SFAC for an image is computed as

$$\text{SFAC} = \frac{\sum_{(i,j)\in \mathbb{P}^k_{\text{bbox}}} \mathbf{I}_k(i, j)}{\sum_{(i,j)\notin \mathbb{P}^k_{\text{bbox}}} \mathbf{I}_k(i, j)} \tag{10}$$

where $\mathbb{P}^k_{\text{bbox}}$ represents the set of pixels inside the ground-truth bounding boxes. The final SFAC score is obtained by averaging over all images in the test set. To further analyze the SFAC across different object sizes, we calculate the metric for different target size categories. Specifically, we categorize targets into extremely small (es), relatively small (rs), generally small (gs), and normal size (nl) targets, corresponding to targets with an area less than or equal to 144 pixels, greater than 144 but less than or equal to 400 pixels, greater than 400 but less than or equal to 1024 pixels, and greater than 1024 pixels, respectively. We also compute the SFAC for all

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TABLE V
QUANTITATIVE COMPARISON OF SFAC METRICS BETWEEN THE BASELINE AND OUR METHOD FOR DIFFERENT TARGET SIZES

Method	SFAC@es	SFAC@rs	SFAC@gs	SFAC@nl	SFAC
w/o MTA&TPA	1.35	1.55	1.78	1.95	1.91
w/ MTA&TPA	1.50(+**0.15**)	1.73(+**0.18**)	2.22(+**0.44**)	2.52(+**0.57**)	2.47(+**0.56**)

target sizes combined to provide a holistic evaluation. The results are shown in Table V.

Quantitative results indicate that our method significantly enhances the model's focus on targets compared to background noise. Specifically, there are notable improvements for generally small targets and normal targets, with increases of 0.44 and 0.57, respectively. The attention to extremely small and relatively small targets also shows improvement. Overall, our method enhances the model's target perception capability by approximately 30%. These quantitative results strongly demonstrate the effectiveness of our method.

V. CONCLUSION

In this work, we proposed DMM, a framework for multispectral-oriented object detection in remote sensing. Our DMM includes a DCFM module, which integrates global context information by leveraging modality disparity information as guidance to adaptively resolve intermodal conflicts, thereby achieving efficient cross-modal feature fusion. Additionally, we designed an MTA module to suppress noise and focus on target regions, minimizing the impact of intramodal variations on subsequent fusion. To ensure the effectiveness of the MTA module, we applied a TPA auxiliary task, using single-modal losses as penalty terms to constrain the optimization process of the model. Extensive experiments on two challenging datasets demonstrate the generalization capability of DMM, achieving SOTA results.

Although DMM demonstrates outstanding performance on multimodal remote sensing images, its current design relies on the premise of consistent spatial resolution across different modalities. However, in remote sensing scenarios, the image resolution across modalities sometimes varies, leading to inconsistencies in image features at the system level. In future research, we plan to explore more effective strategies for fusing multimodal information under varying resolution scales to address this challenge.

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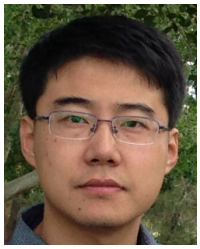
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